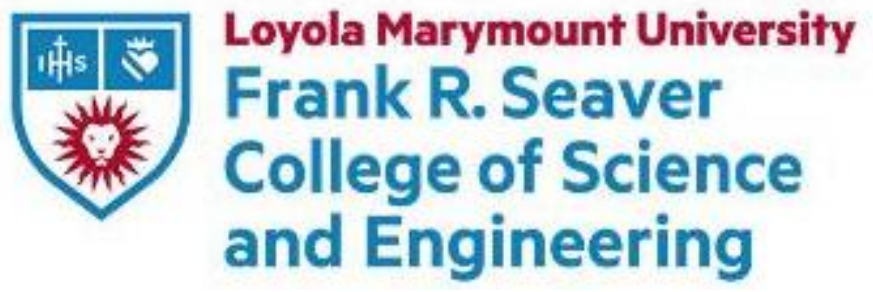


Sensitivity analysis of GRNmap and new features for GRNsight: open source software for dynamical systems modeling and visualization of small-scale gene regulatory networks

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<http://kdahlquist.github.io/GRNmap/> and <http://dondi.github.io/GRNsight/>



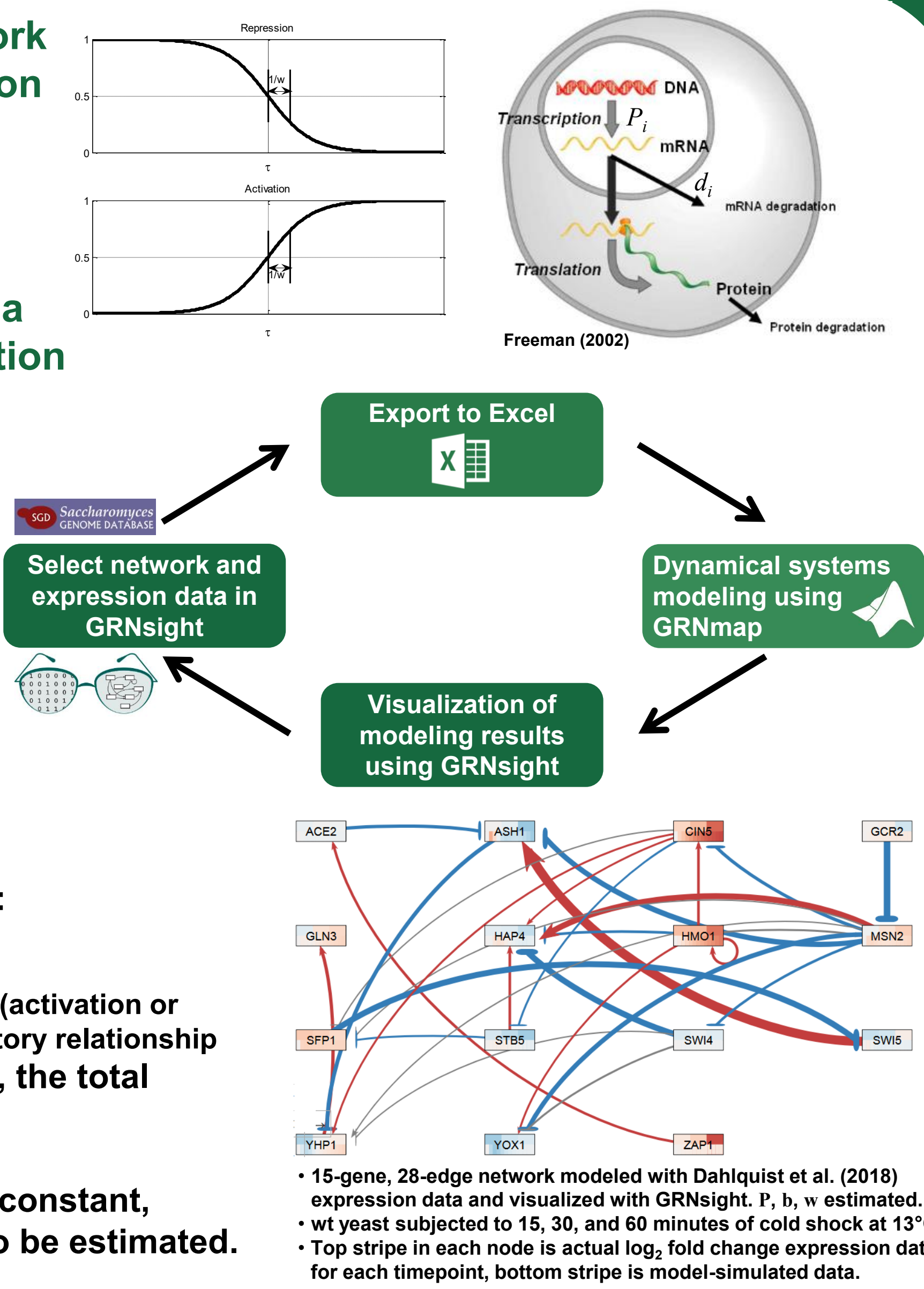
GRNmap: Gene Regulatory Network modeling and parameter estimation

$$\frac{dx_i(t)}{dt} = \frac{P_i}{1 + \exp\left(-\left(\sum_j (w_{ij}(t) - b_j)\right)\right)} - d_i x_i(t)$$

Parameters are optimized using a penalized least squares error function

$$E = \alpha \|\theta\|^2 + \frac{1}{Q} \sum_{t=1}^Q [z^d(t_r) - z^e(t_r)]^2$$

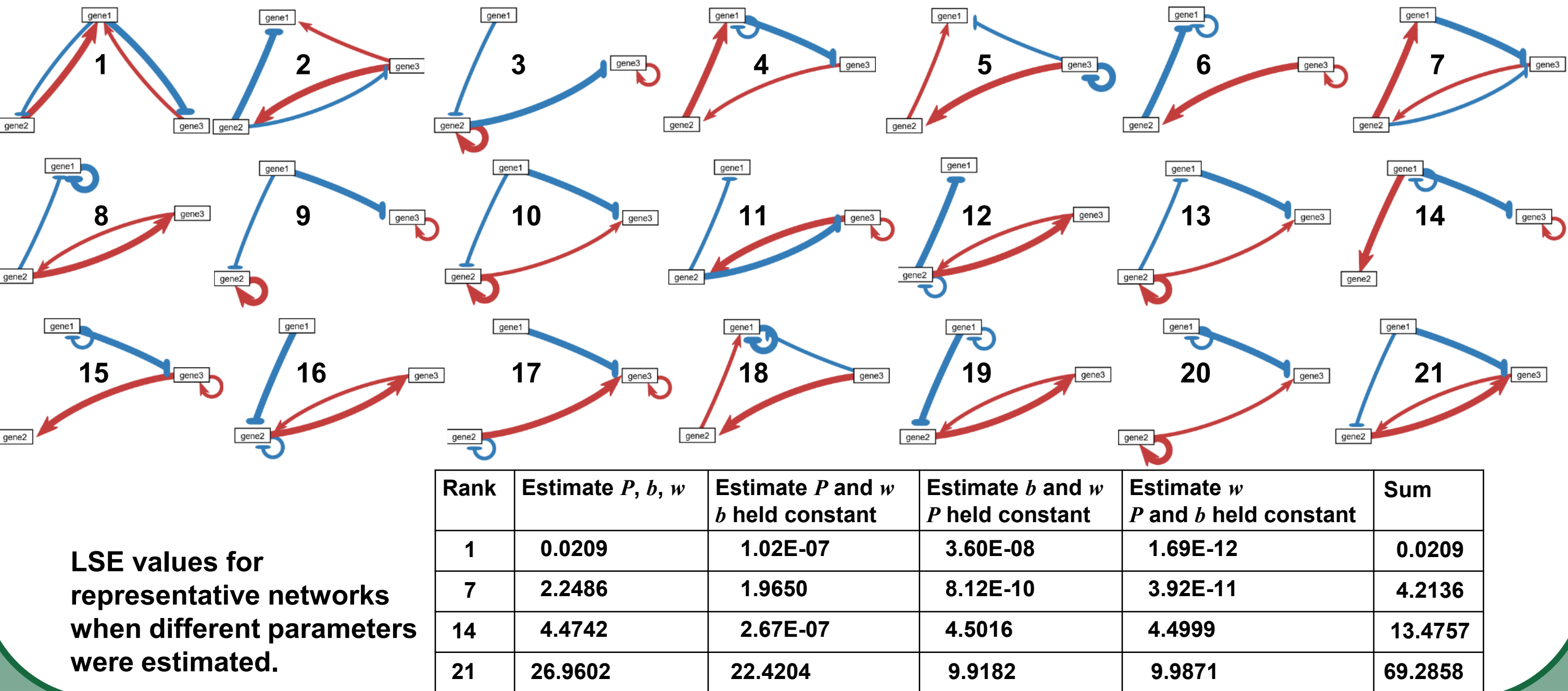
- Each node represents one gene encoding a transcription factor.
- Each edge represents a regulatory relationship.
- Model is coded in MATLAB.
- Input Excel workbook requires:
 - Network adjacency matrix
 - Timecourse expression data, $x(t)$
 - mRNA degradation rates, d
- Output workbook provides optimized:
 - Production rates, P
 - Expression threshold, b
 - Weight parameters, w , give the direction (activation or repression) and magnitude of the regulatory relationship
- When P s, b s, and w s are all estimated, the total number of parameters is (2 X no. of genes) + no. of edges.
- However, P , b , or P and b can be held constant, reducing the number of parameters to be estimated.



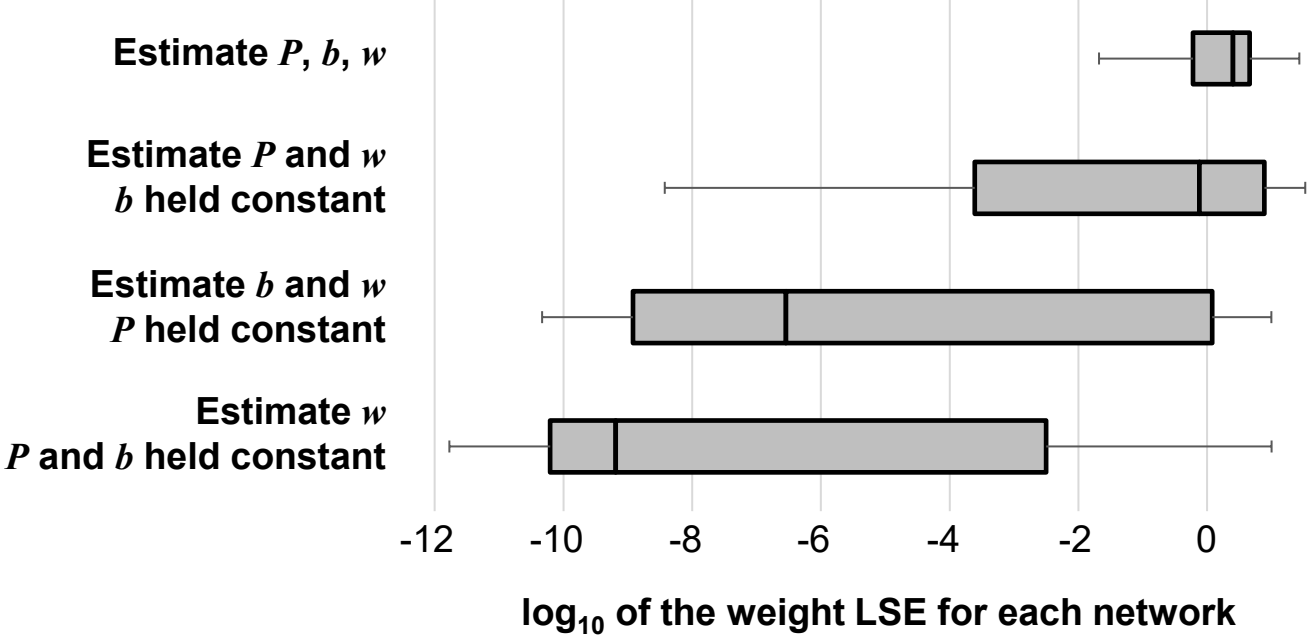
Model sensitivity was tested using all 21 possible 3 gene, 4 edge “toy” networks. Simulated data was generated based on known arbitrary parameters. Then the simulated data was used to estimate parameters to see if the model returned the known values.

- Least squares error (LSE) was used to rank the performance of the networks relative to one another.
- The error between the known weights and the estimated weights was computed as follows:

$$LSE = \sum_{i=1}^n (w_i - w'_i)^2 \text{ where } w_i \text{ are the actual weights and } w'_i \text{ are the estimated weights.}$$



Weight estimation precision increases when production rates are held constant

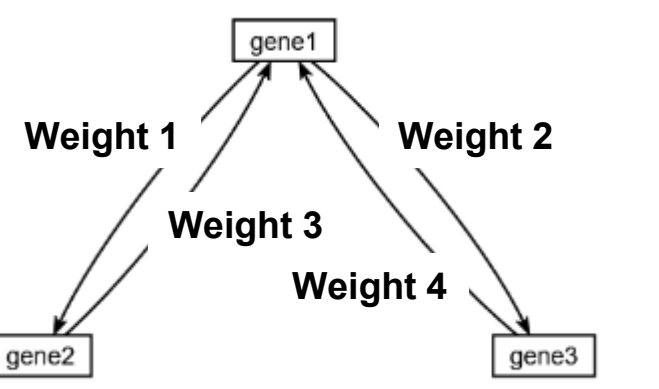


- When P and b were held constant, only the 4 weight parameters were estimated, compared to the 10 estimated when P, b , and w were all estimated.
- The performance of the model when P and b were held constant and when only P was held constant were relatively similar, suggesting that the issue is estimating the production rate, not necessarily the number of parameters estimated.
- We did not discern a clear pattern in the network structures that corresponded to model performance.

Weight estimation precision is dependent on weight values

- Since weights were chosen arbitrarily (1, -1, 2, or -2), we also investigated whether the model performance was solely based on network structure, or whether it was also impacted by the assignment of a particular weight value to each edge.
- All 24 permutations of the weight assignments to edges were modeled for the networks originally ranked 1st and 7th to determine the impact of the weights.
- The LSE of the weights were calculated for each trial as before.

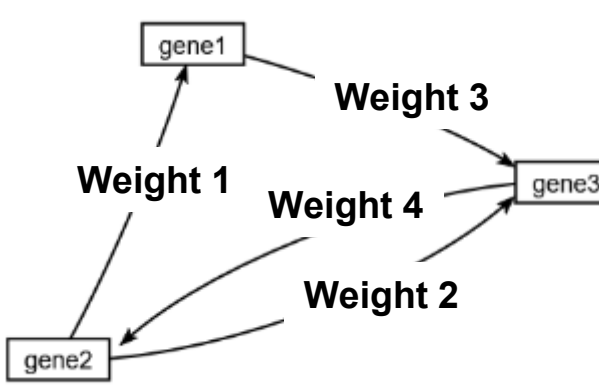
† Network Originally Ranked 1st



*Best	Actual Weights	Weight estimates with P and b held constant	Weight estimates when P and b also estimated
Weight 1	-2	-2.0000	-1.9172
Weight 2	-1	-1.0000	-0.9030
Weight 3	1	1.0000	0.9764
Weight 4	2	2.0000	1.9388

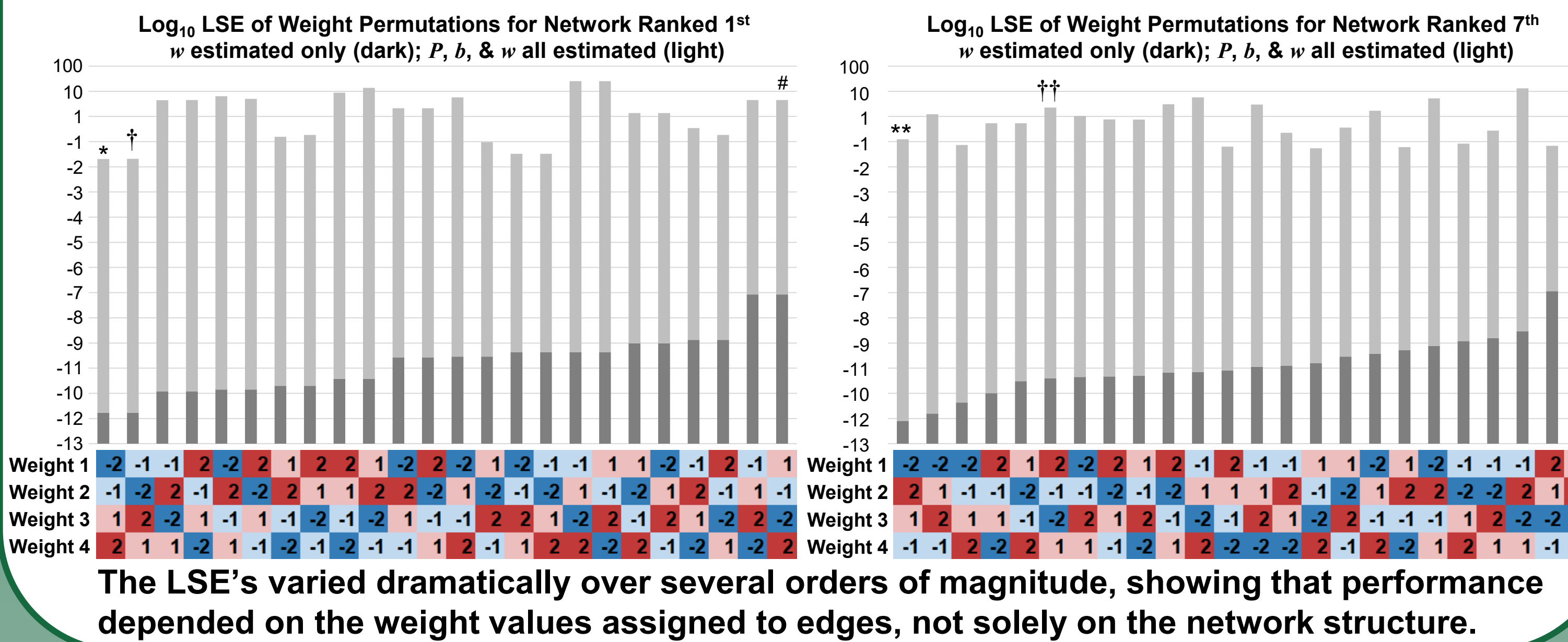
**Worst	Actual Weights	Weight estimates with P and b held constant	Weight estimates when P and b also estimated
Weight 1	1	1.0000	0.4924
Weight 2	-1	-1.0000	-0.7337
Weight 3	-2	-1.9999	-2.8069
Weight 4	2	1.9997	0.1271

†† Network Originally Ranked 7th



**Best	Actual Weights	Weight estimates with P and b held constant	Weight estimates when P and b also estimated
Weight 1	-2	-2.0000	-2.0121
Weight 2	2	1.9999	2.3139
Weight 3	1	1.0000	1.1456
Weight 4	-1	-1.0000	-0.9312

##Worst	Actual Weights	Weight estimates with P and b held constant	Weight estimates when P and b also estimated
Weight 1	1	1.0000	0.6198
Weight 2	2	1.9993	1.5623
Weight 3	-2	-1.9996	-1.5411
Weight 4	-1	-1.0000	-0.7627



GRNsight, automatically lays out a gene regulatory network (GRN) graph, coloring the edges based on regulatory weights, and the nodes based on experimental or simulated expression data

1. Network

- Demo files are provided.
- Can import Excel, SIF, or GraphML files.
- Can construct a network from SGD regulation or PPI data stored in a backend database.

2. Layout

- Users can select force graph or grid layout.
- Force graph parameter sliders:
 - Link distance determines the minimum distance between nodes.
 - Nodes have a charge which repels or attracts other nodes.

3. Node

- Coloring can be based on user-uploaded expression data or from published expression datasets stored in a backend database.
- Datasets available:
 - Apweiler et al. 2012, GSE33098
 - Barreto et al. 2012, GSE24712
 - Dahlquist et al. 2018, GSE83656
 - Kitagawa et al. 2002, GSE9336
 - Thorsen et al. 2007, GSE6068

4. Edge

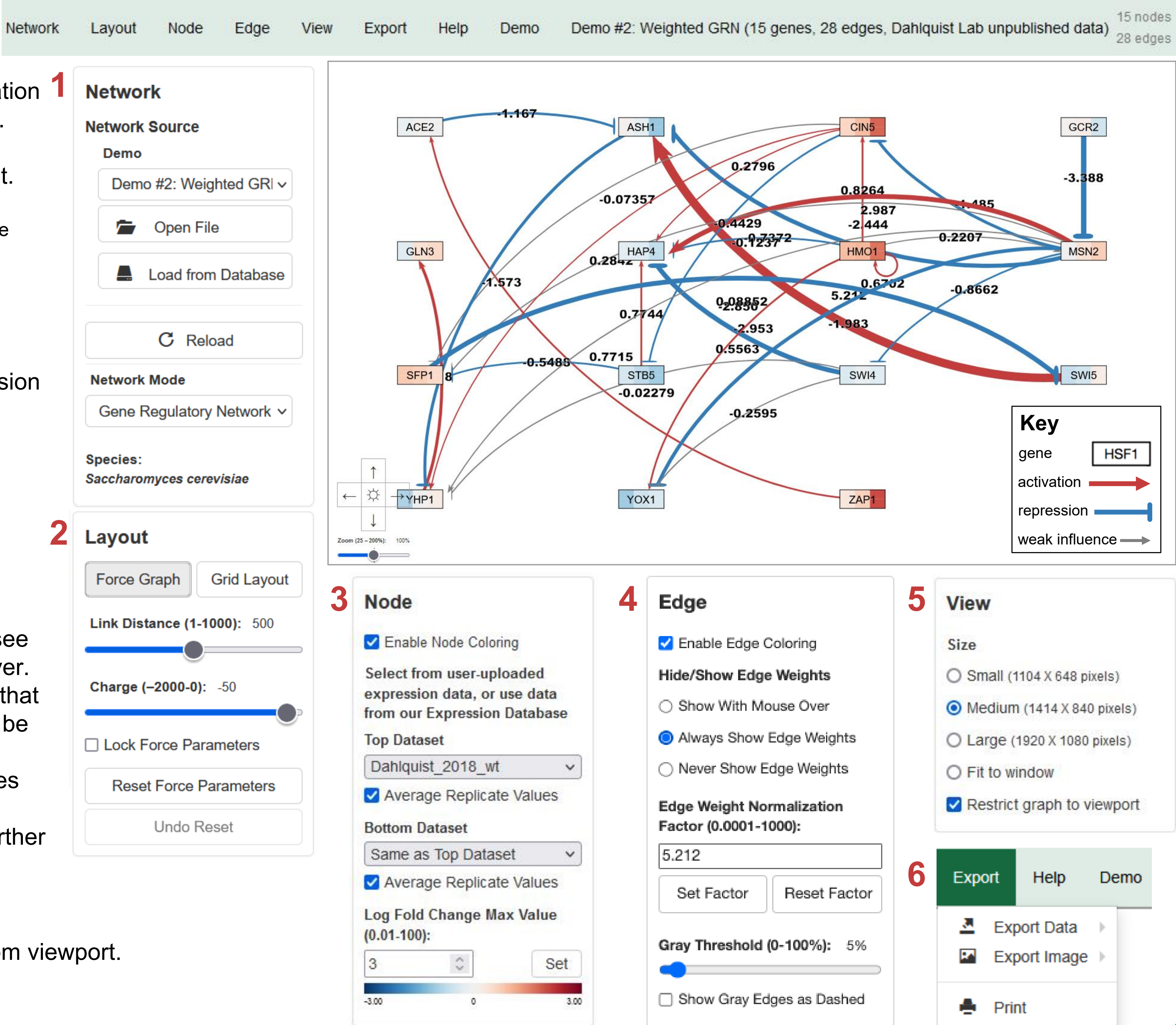
- Edge coloring can be toggled on and off.
- Buttons enable the user to always/never see edge weights, or see them upon mouseover.
- Allows user to set normalization factor so that edge thicknesses for different graphs can be rendered on the same scale.
- Changing the gray threshold deemphasizes weaker edges in the visualization.
- Presenting gray edges as dashed lines further deemphasizes weak edges.

5. View

- Multiple viewport sizes available.
- Graph bounding box can be separated from viewport.
- Zooming and scrolling available.

6. Export

- Basic data export to SIF or GraphML or full GRNmap-compatible Excel workbook, with yeast mRNA degradation rates from Neymotin et al. (2014).
- Image export to PNG, SVG, or PDF.



Conclusions

- Estimating production rates from gene expression data is difficult for GRNmap to do, and we plan to determine production rates from experiments.
- Investigation of 3-gene, 4-edge “toy” networks shows that the performance of the GRNmap model is influenced by a combination of network structure and edge weights.
- The GRNmap – GRNsight system will speed up the rate of modeling and analysis of many networks.

Availability

- GRNmap and GRNsight are free and open to all users and there is no login requirement.
 - <http://kdahlquist.github.io/GRNmap/>
 - <http://dondi.github.io/GRNsight/>
- GRNmap and GRNsight code are available under the open source BSD license at our GitHub repositories.
 - <https://github.com/kdahlquist/GRNmap/>
 - <https://github.com/dondi/GRNsight>
- The GRNmap model can be run from code only on Windows with MATLAB versions 2014b or 2024b.
- Although originally developed with yeast data, both GRNmap and GRNsight can be used with any species for which you have time course gene expression data or an unweighted or weighted adjacency matrix, respectively.

Acknowledgments

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